

California Employment & Wage Trends

(2020–2025 Analysis)

Presented By

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Occupational Employment & Wage Statistics (OEWS)

- **Source:** U.S. Bureau of Labor Statistics. (n.d.). Occupational Employment and Wage Statistics (OEWS) [Data set]. Data.gov.
- **Time Period:** 2020 - 2025
- 22 Occupation Groups

Occupational Employment & Wage Statistics (OEWS)

Fields			
Type	Field Name	Physical Table	Remote Field Name
	County ▼	Sheet1	County
Abc	Area Name ▼	Sheet1	Area Name
	Year ▼	Sheet1	Year
Abc	Occupation Group ▼	Sheet1	Occupation Group
Abc	Occupational Title ▼	Sheet1	Occupational Title
Abc	Wage Type ▼	Sheet1	Wage Type
	Number Of Employed ▼	Sheet1	Number Of Employed
	Wage ▼	Sheet1	Wage



Research Question 1

**Do high-growth occupations
experience a wage increase
between 2020 and 2025?**





Why this matters for us (students)

- We're all graduating soon — so **knowing which jobs are actually growing** helps us *avoid dead-end careers*.
 - If high-growth roles also pay more, that's **key for choosing career paths with real earning potential**.
 - **Understanding these trends helps us plan smarter** — for salaries, job security, and long-term opportunities.
- 



Why this matters for decision-makers

- Highlights where **talent shortages** are emerging.
- Helps **predict wage pressure** and **hiring costs**.
- Informs **workforce planning across** different **occupation groups**.





Evidence

- **Healthcare Support Occupations are projected to grow ~15%** nationally (2022–2032) — high demand should put upward pressure on wages. (Bureau of Labor Statistics, 2023)
 - **STEM Occupations are projected to grow 8–10% and already command higher median wages than non-STEM fields** — strong demand and specialized skills typically drive wage increases. (Bureau of Labor Statistics, 2025)
- 



H1A

**As employment increases across
occupations, wages increase to
support this demand.**



Calculated Fields

BaseYearEmployment

```
{ FIXED [Occupation Group] :  
  AVG( IF [Year] = 2020 THEN [Number Of Employed] END )  
}
```

BaseYearWage

```
{ FIXED [Occupation Group] :  
  AVG( IF [Year] = 2020 THEN [Wage] END )  
}
```

This formula returns the **2020 base-year average # of employed** and **wage** (respectively) for each **Occupation Group**, and that value stays **constant** for the group no matter what other filters or dimensions you have in the view.





Calculated Fields

NormEmployment

```
( AVG ( [Number Of Employed] ) - MIN ( [BaseYearEmployment] ) )  
/ MIN ( [BaseYearEmployment] )
```

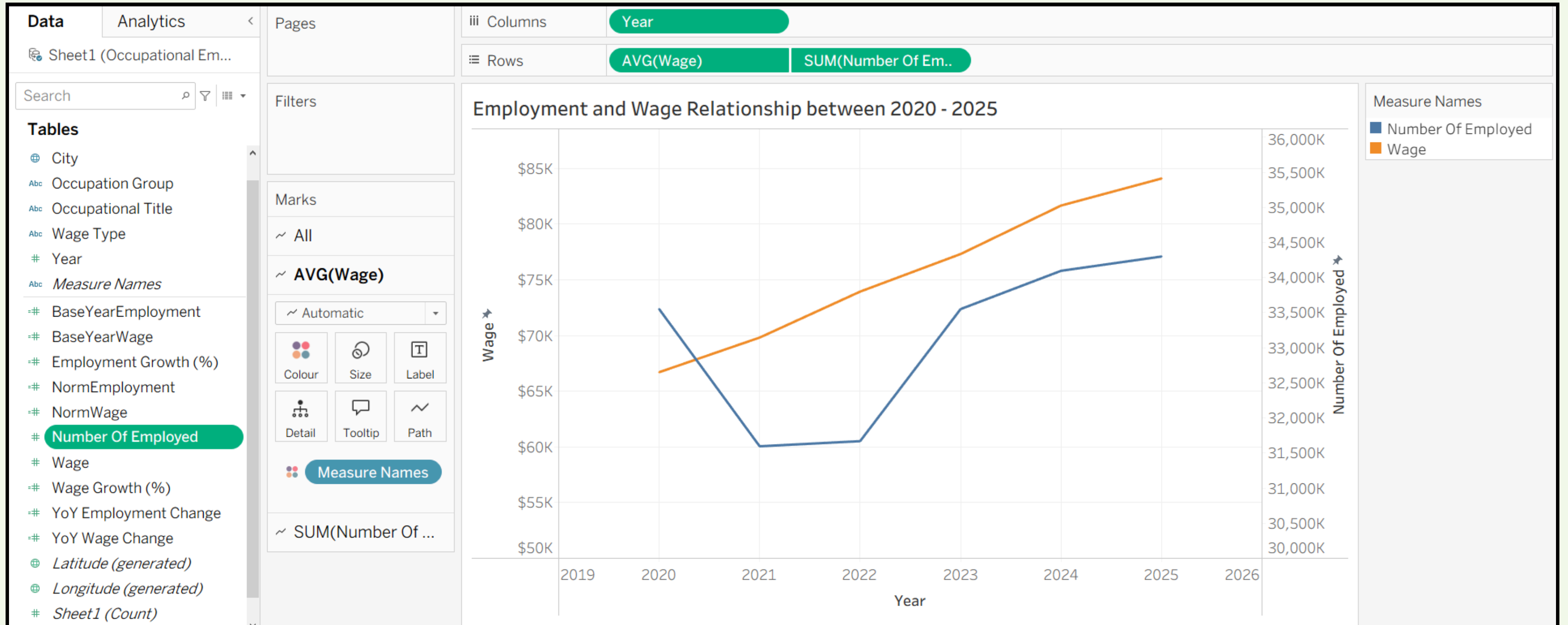
This formula gives you the percent **change in employment relative to the base year** (2020) for each occupation group.

NormWage

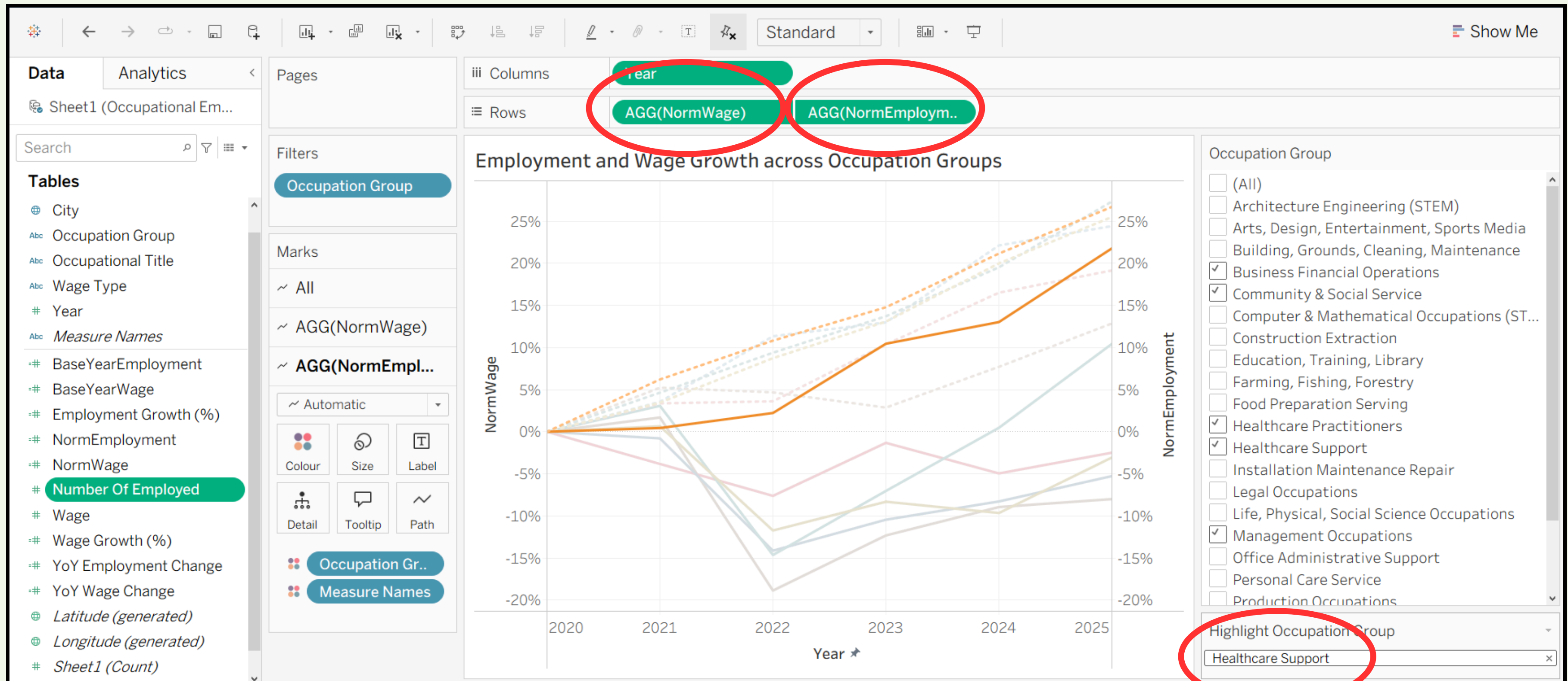
```
( AVG ( [Wage] ) - MIN ( [BaseYearWage] ) )  
/  
MIN ( [BaseYearWage] )
```

This formula gives you the percent **change in wage relative to the base year** (2020) for each occupation group.

Visualization H1A



Visualization H1A

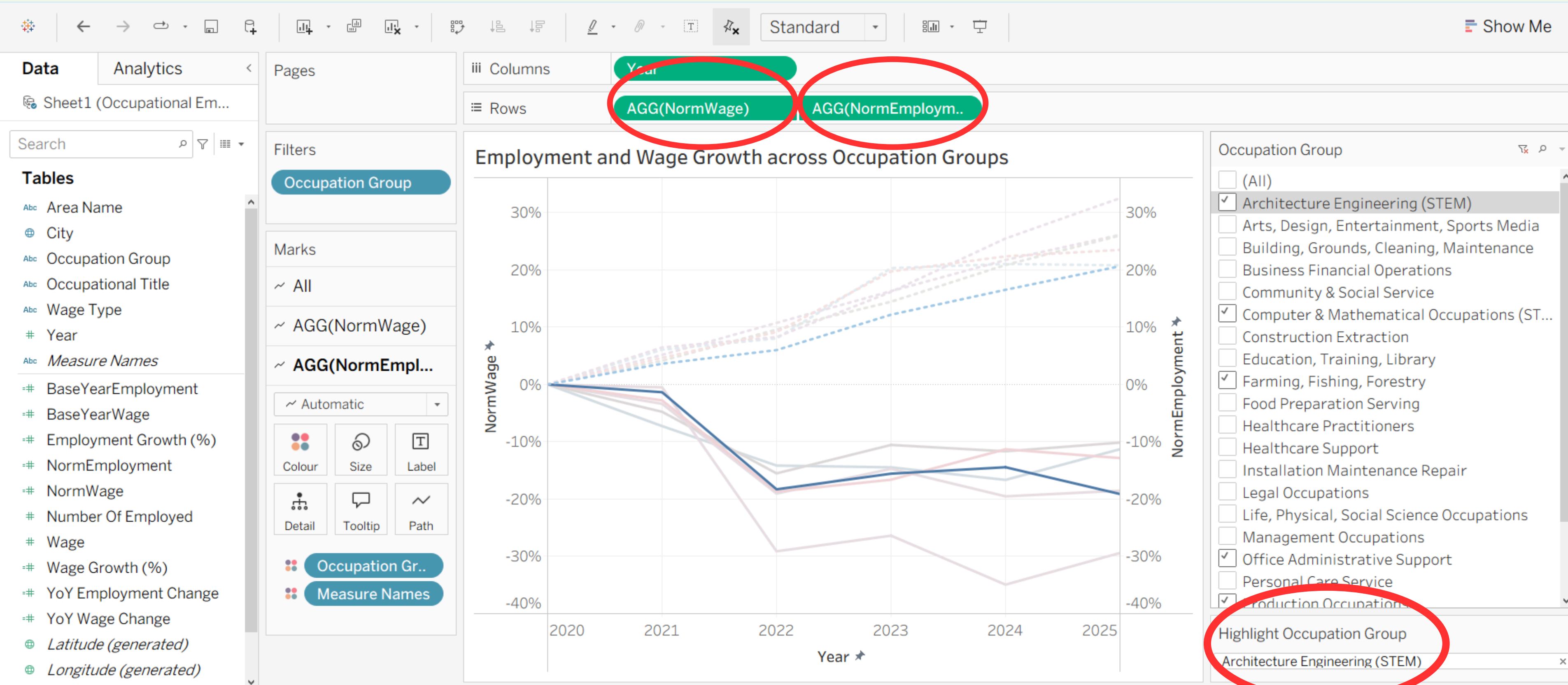


Evidence Supporting the Hypothesis

*As employment increases across occupations,
wages increase to support this demand.*

- The COVID-19 period (2020–2022) creates significant **volatility** in both employment and wages, disrupting typical demand–wage relationships.
- **Wages rise as employment grows** from 2023–2025, showing a positive demand–wage relationship in several occupations (e.g., Business Financial Operations, Community & Social Service, Healthcare).
- **High-demand occupations** demonstrate **stronger wage responsiveness**.

Visualization H1A



Evidence Contradicting the Hypothesis

*As employment increases across occupations,
wages increase to support this demand.*

- Many occupation groups (Architecture Engineering, Computer and Mathematical Occupations, Farming), **do not show parallel growth** between employment and wages from 2020–2025.
- While employment levels often decline or recover slowly after 2022, wage patterns **do not consistently move in the same direction.**



Managerial Recommendations

- Prepare for **rising wage pressure** in high-growth occupations.
 - Adjust compensation strategies to stay competitive.
- Strengthen **retention efforts**.
 - Improve benefits, training, and flexibility.
- **Invest in fast-growing occupation groups** (e.g., STEM, Business, etc.)
 - Prioritize recruitment and upskilling.
- **Use non-wage incentives** where wages lag behind demand.
 - Offer flexibility, development paths, and workplace benefits.





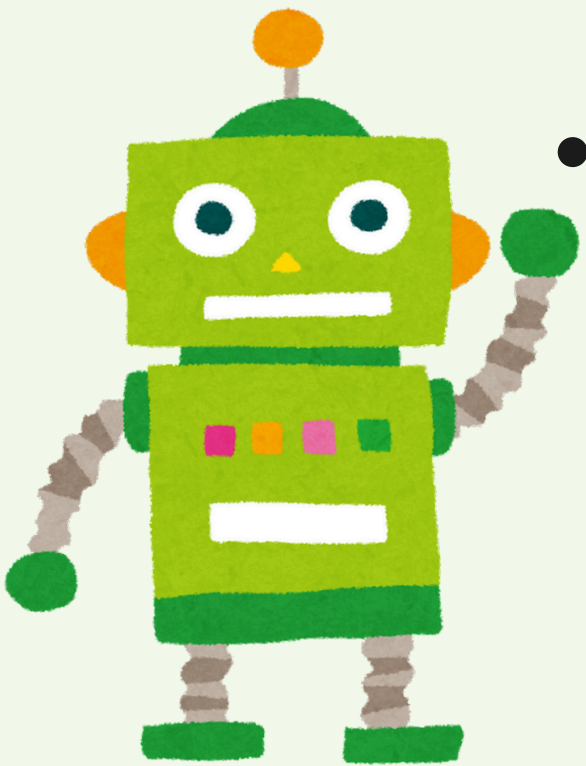
Claude

Opus 4.5



Visualization with AI

- **Research Question:** Do high-growth occupations experience corresponding wage increase between 2020 and 2025?
- **Hypothesis:** As employment increases across occupations, wages increase to support this demand.
- **Prompt:** Use the Dataset attached and create a visualization to explain whether you accept or reject the hypothesis 1A and why.



Employment vs. Wage Trends in California (2020-2025)

Market Research: Hypothesis 1A Analysis

Hypothesis 1A — Positive Demand-Wage Relationship

Statement: As employment increases across occupations, wages increase to support this demand.

+2.2%

Employment Growth
(2020-2025)

+24.9%

Wage Growth
(2020-2025)

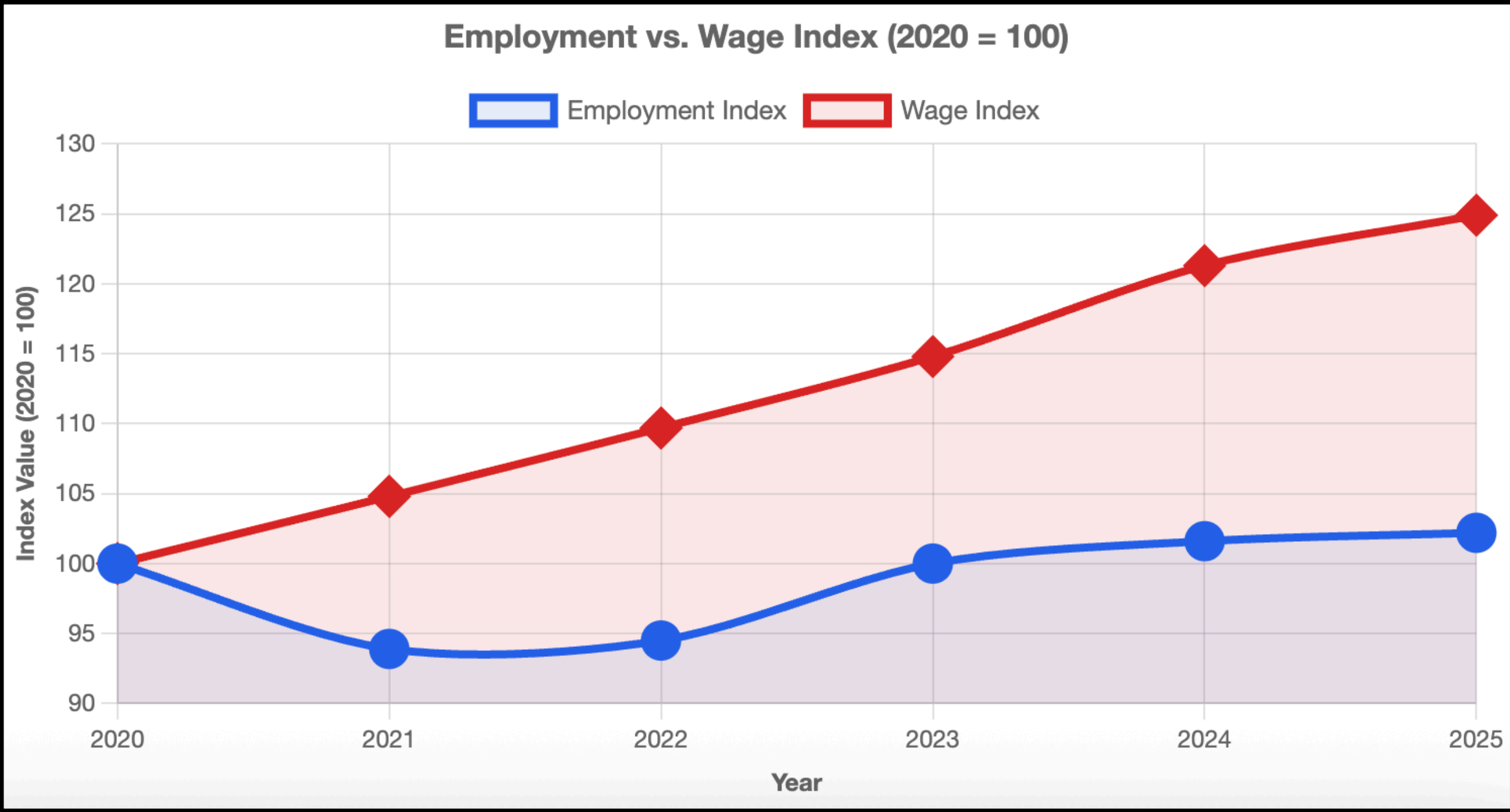
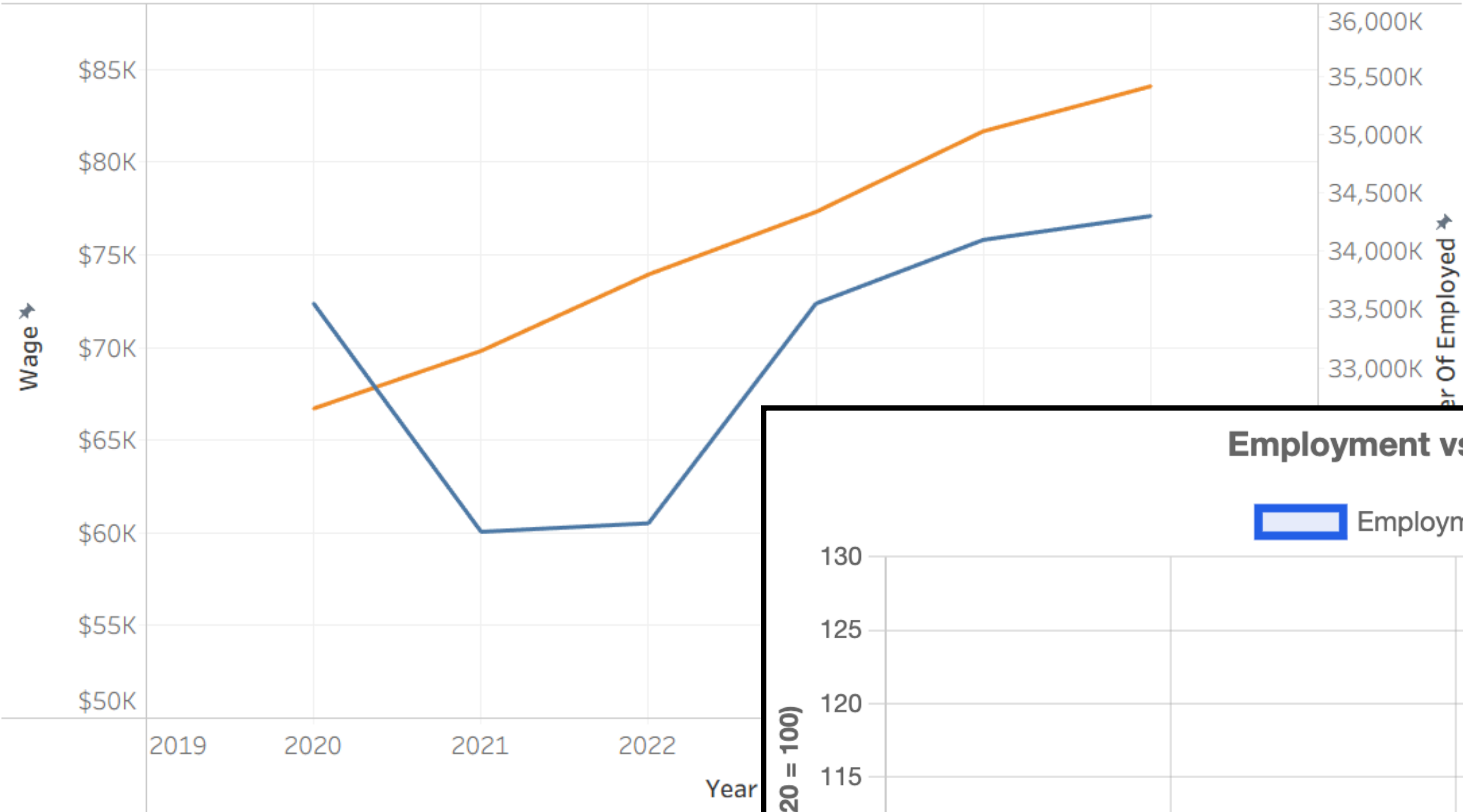
$r = 0.608$

Correlation
Coefficient

Employment vs. Wage Index (2020 = 100)



Employment and Wage Relationship between 2020 - 2025





Evidence

- Workers in **occupations with non-routine analytical & interpersonal skills experience the highest wage growth** — not all occupations benefit equally. (Abraham et al., 2024)
 - The U.S. Bureau of Labor Statistics reports that the fastest-growing jobs — especially in tech and healthcare — often **show strong wage gains as demand for specialized skills rises**. (Bureau of Labor Statistics, 2025)
- 

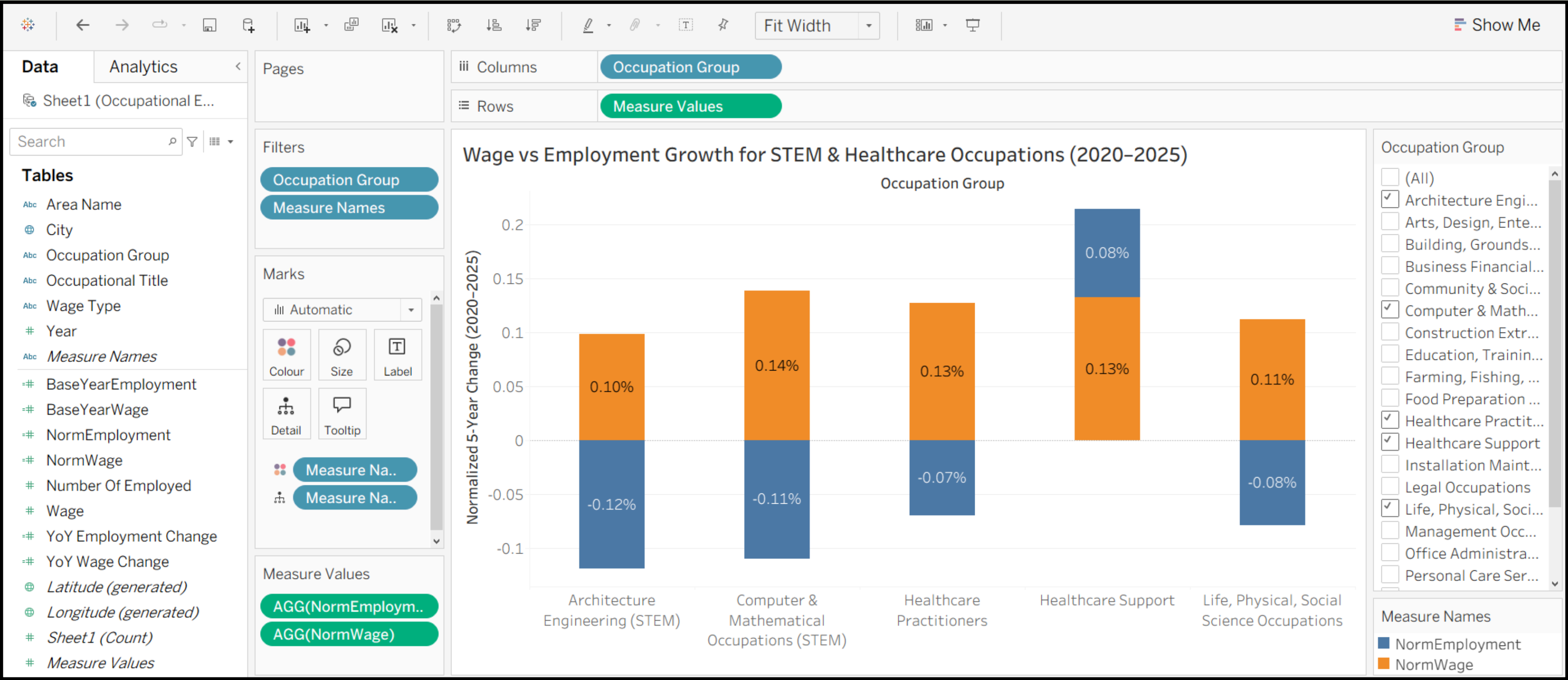


H1B

Only certain fast-growing fields, like STEM and healthcare, experience noticeable wage increases when employment demand goes up.



Visualization H1B



Evidence Supporting the Hypothesis

Only certain fast-growing fields, like STEM and healthcare, experience noticeable wage increases when employment demand goes up.

- Within our selected STEM and Healthcare groups, only **one occupation** matches the pattern predicted by the hypothesis.
- Healthcare Support shows **both rising** normalized employment and rising normalized wages.
- This occupation demonstrates that in certain high-skill fields, increased **demand for specialized labor** can translate into wage growth.

Evidence Contradicting the Hypothesis

Only certain fast-growing fields, like STEM and healthcare, experience noticeable wage increases when employment demand goes up.

- Most occupations **do not indicate a direct correlation** between wage and employment increases.
- Several occupations — such as Computer & Mathematical, Architecture Engineering, Life/Physical/Social Science — show wage growth despite declining employment.
- These patterns **contradict** the hypothesis by showing that wage gains can occur even in occupations without rising demand.



Managerial Recommendations

- Focus **recruitment and retention** efforts on occupations where both wages and employment are rising.
- Use targeted **upskilling and role redesign** for fields with rising wages but declining employment.
- Base workforce planning on **both wage and employment trends** rather than assuming demand drives pay.





Research Question 2

Has wage growth kept pace with employment growth in high growth regions, compared to the low growth regions between 2024 and 2025?





Why this matters for decision-makers

- Wage-employment differences across regions affect **hiring competitiveness and talent retention**.
- Regions with rising employment but lower wages may face **recruitment pressure**.
- Comparing regions in the **most recent year** gives a clear snapshot for compensation and workplace planning.





Evidence

- **High-growth U.S. metros** show **stronger employment momentum** and also **elevated wage levels**, reflecting business expansion and intensified competition for skilled workers. (CBRE, 2025)
 - Many **fast-growing metropolitan regions** in the U.S. also **report higher-than-average wages**, especially in markets with active tech, healthcare, or professional service sectors. (Bureau of Labor Statistics, 2023)
- 

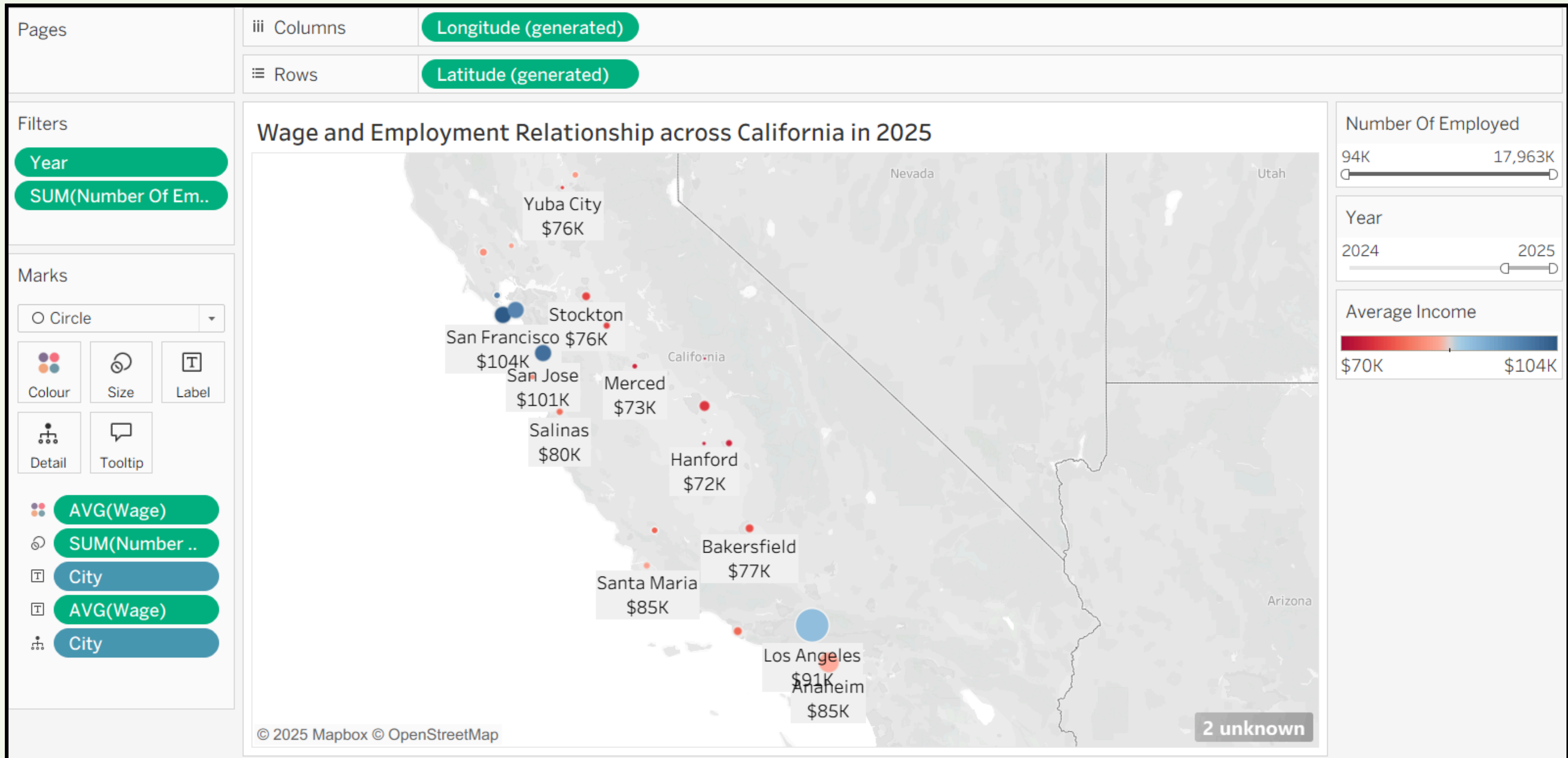


H2

Regions experiencing stronger employment growth will also exhibit higher average wages.



Visualization H2





Conclusion H2

Regions experiencing stronger employment growth will also exhibit higher average wages.

- **Regions with greater labor demand** offer **higher wages** to attract and retain workers.
 - **Regions with lower labor demand** maintain **lower wage levels** due to weaker competition for talent.
- **The data supports the hypothesis that wages rise in cities with larger employment bases and higher labor-market competition.**
- 



Managerial Recommendations

- **Invest in talent development in low-employment regions** to raise competitiveness.
 - **Promote inland regions' lower costs** to attract businesses and jobs.
 - **Strengthen retention efforts in high-employment metros** (benefits, flexibility, career pathways).
- 

Tableau Dashboard

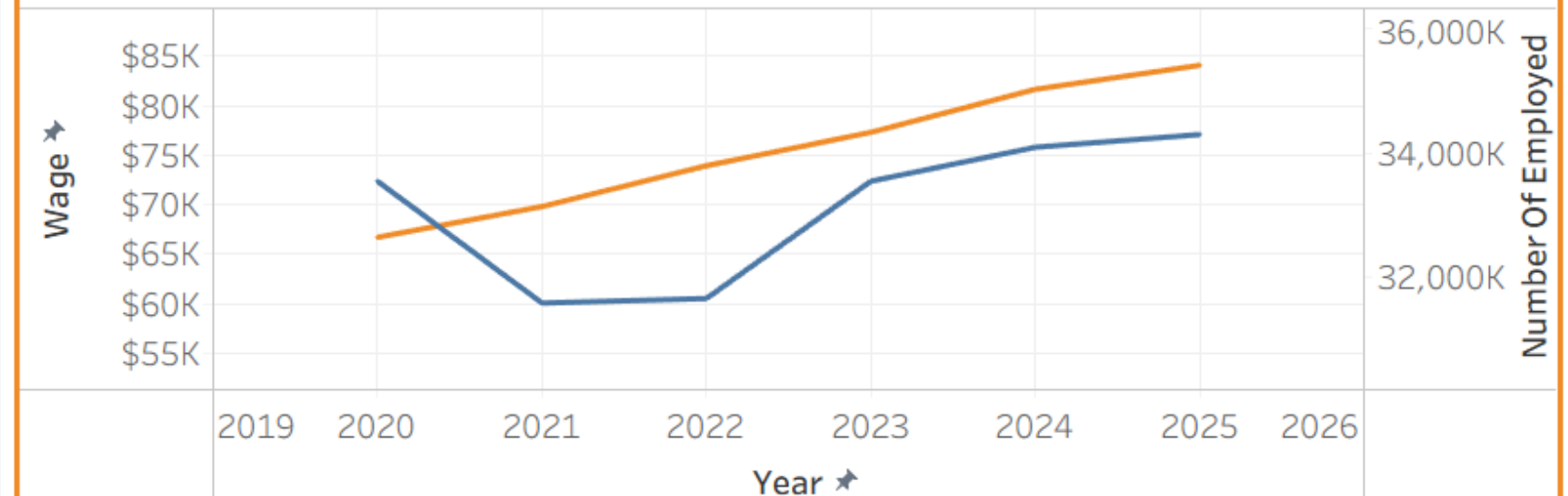
Wage and Employment Relationship across California in 2025



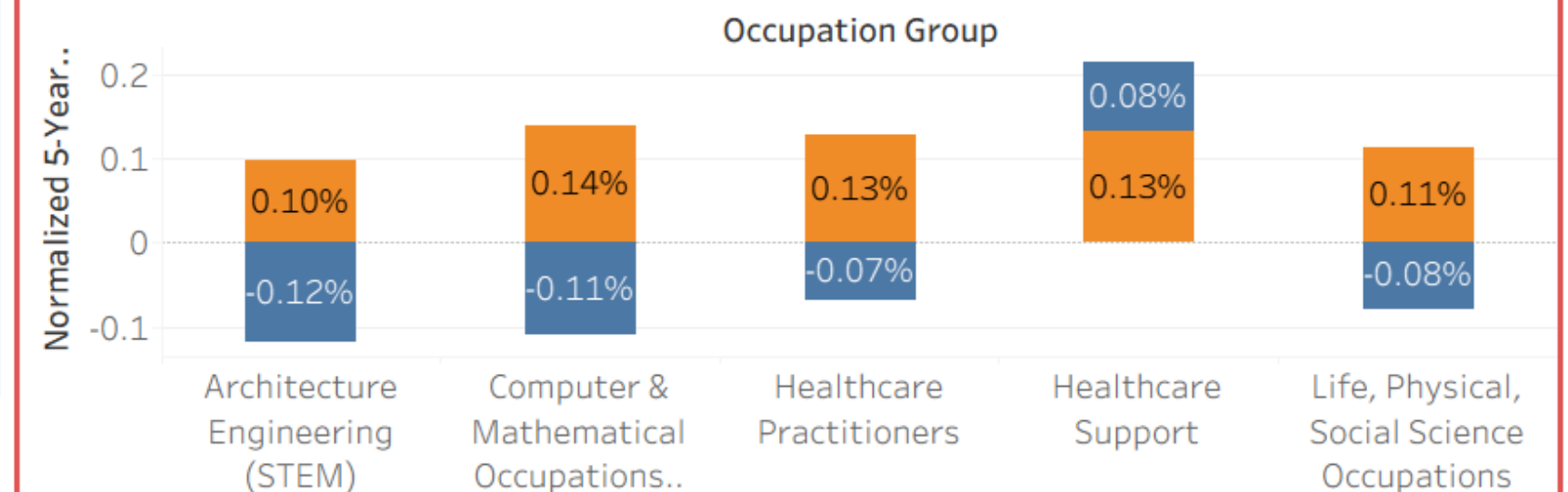
Average Income

\$70K \$104K

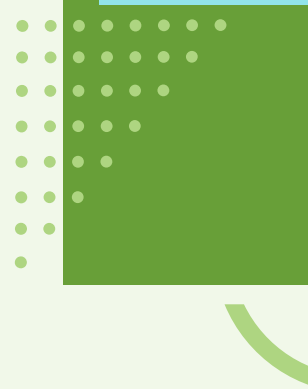
Employment and Wage Relationship between 2020 - 2025



Wage vs Employment Growth for STEM & Healthcare Occupations (2020-2025)



Big picture: it all comes down to **demand**—fields that are growing shape both how many jobs exist and how competitive wages need to be, making it crucial to **invest in high-growth areas** if we want to attract and keep talent.



THANK YOU

END



Reference List

- Abraham, L., Mulhern, C., Oppen, I. M., Puri, P., & Roth, E. (2024). Wage growth and initial occupations: Examining differences across worker demographics using longitudinal data. RAND Corporation. https://www.rand.org/pubs/research_briefs/RBA3492-1.html
- Bureau of Labor Statistics. (2025). Employment in STEM occupations. U.S. Department of Labor. <https://www.bls.gov/emp/tables/stem-employment.htm>
- Bureau of Labor Statistics. (2025). Fastest-growing occupations, 2024–34. U.S. Department of Labor. <https://www.bls.gov/ooh/fastest-growing.htm>
- Bureau of Labor Statistics. (2023). Employment projections: 2022–2032 summary. U.S. Department of Labor. https://www.bls.gov/news.release/archives/ecopro_09062023.pdf
- CBRE. (2025). Where is job growth accelerating? Shifting employment momentum in top metros. CBRE Research. <https://www.cbre.com/insights/briefs/where-is-job-growth-accelerating-shifting-employment-momentum-in-top-metros>
- U.S. Bureau of Labor Statistics. (n.d.). Occupational Employment and Wage Statistics (OEWS) [Data set]. <https://catalog.data.gov/dataset/occupational-employment-and-wage-statistics-oevs-4b4c4>



Tableau Visualization Step-by-Step Explanations



Calculated Fields

BaseYearEmployment

```
{ FIXED [Occupation Group] :  
    AVG( IF [Year] = 2020 THEN [Number Of Employed] END )  
}
```

{ FIXED [Occupation Group] : ... }

Tableau to compute the expression inside once per [Occupation Group], ignoring other dimensions in the view.

IF [Year] = 2020 THEN [# of Employed] END

This conditional returns the # of employed value only for the year 2020. For any other year, it returns NULL.

AVG(...)

Tableau takes the average of the returned wages for 2020 within each Occupation Group.

BaseYearWage

```
{ FIXED [Occupation Group] :  
    AVG( IF [Year] = 2020 THEN [Wage] END )  
}
```

{ FIXED [Occupation Group] : ... }

Tableau to compute the expression inside once per [Occupation Group], ignoring other dimensions in the view.

IF [Year] = 2020 THEN [Wage] END

This conditional returns the Wage value only for the year 2020. For any other year, it returns NULL.

AVG(...)

Tableau takes the average of the returned wages for 2020 within each Occupation Group.

This formula returns the **2020 base-year average # of employed** and **wage** (respectively) for each Occupation Group, and that value stays **constant** for the group no matter what other filters or dimensions you have in the view.

Calculated Fields

NormEmployment

```
( AVG([Number Of Employed]) - MIN([BaseYearEmployment]) )  
/ MIN([BaseYearEmployment])
```

AVG([Wage])

This gives the average # of employed for the current row's level of detail—typically for a specific year and occupation group.

MIN([BaseYearEmployment])

Using MIN() here simply returns that same fixed value, because there is only one BaseYearEmployment per Occupation Group.

AVG([Wage]) - MIN([BaseYearEmployment])

This computes the difference between the current year's # of employed and the base-year (2020) wage. This tells you how much # of employed have increased or decreased since the base year.

Dividing by MIN([BaseYearEmployment])

This converts the difference into a relative change.

NormWage

```
( AVG([Wage]) - MIN([BaseYearWage]) )  
/  
MIN([BaseYearWage])
```

AVG([Wage])

This gives the average wage for the current row's level of detail—typically for a specific year and occupation group.

MIN([BaseYearWage])

Using MIN() here simply returns that same fixed value, because there is only one BaseYearWage per Occupation Group.

AVG([Wage]) - MIN([BaseYearWage])

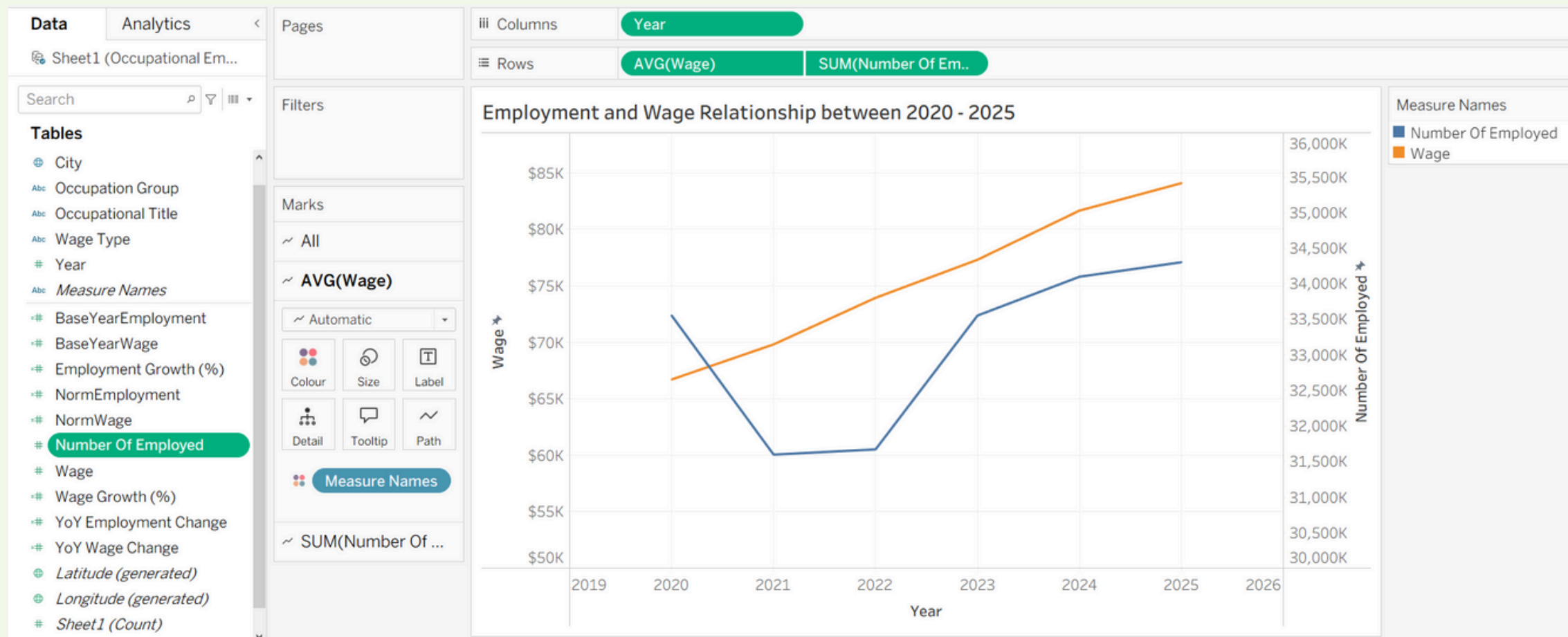
This computes the difference between the current year's wage and the base-year (2020) wage. This tells you how much wages have increased or decreased since the base year.

Dividing by MIN([BaseYearWage])

This converts the difference into a relative change.

This formula gives you the percent **change in employment/wage**, respectively, **relative to the base year** (2020) for each occupation group.

Visualization H1A



1. Columns/Rows:

- Columns: Year (2020–2025 timeline)
- Rows: AVG(Wage) and SUM(Number of Employed)

2. Dual-Axis Setup:

- Employment set to Dual Axis and synchronized.
- Wage formatted as currency
- Employment axis formatted in thousands (K).

3. Marks Card:

- Line mark type applied to both measures.
- Measure Names used for color (Wage = orange, Employment = blue).
- Tooltips customized for clear wage and employment display.

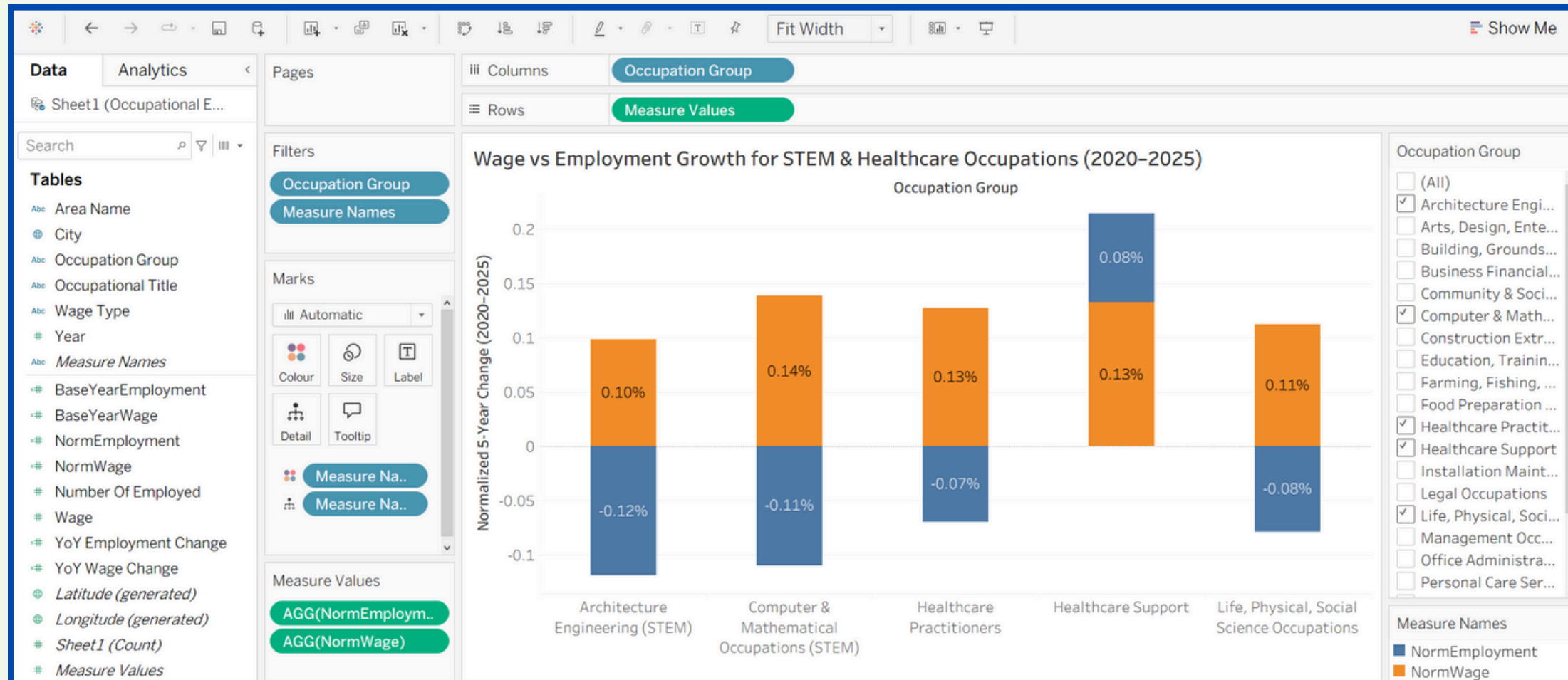
4. Filtering & Sorting:

- Year filtered to 2020–2025.
- Timeline automatically sorted chronologically.

5. Formatting:

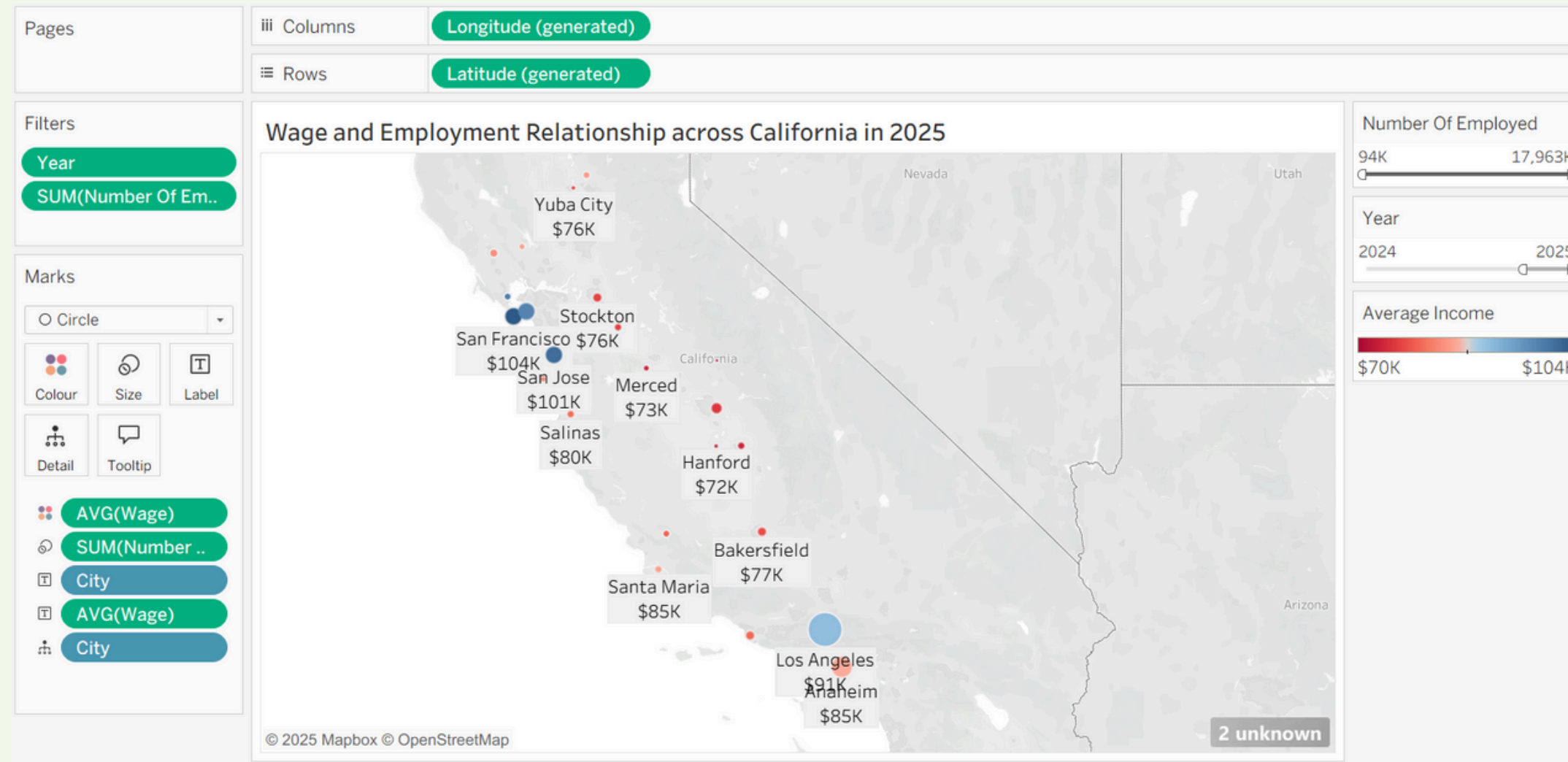
- Title added: “Employment and Wage Relationship Between 2020–2025.”

Visualization H1B



1. Ceated a few calculated fields, for this visualization, we used the two normalized fields, NormEmployment and NormWage, using the formula: $(\text{Average Value} - \text{Base Year Value}) \div \text{Base Year Value}$.
2. This converts five-year changes into a proportional, comparable scale across occupations.
3. We filtered the dataset to focus only on STEM and Healthcare occupations, since these groups directly relate to the research hypothesis.
4. In Tableau, we placed Occupation Group on Columns and added Measure Values to Rows, selecting only NormEmployment and NormWage for display.
5. We used a stacked bar style where positive NormWage values appear above zero and negative NormEmployment values appear below zero. This highlights whether wage and employment shifts move together or diverge.
6. There are distinct color assignments—orange for NormWage and blue for NormEmployment—help make the two metrics immediately distinguishable.
7. Labels were formatted to two decimal places, the Y-axis title was updated to “Normalized 5-Year Change (2020–2025),” and a reference line at zero was added to separate growth from decline.
8. The final visualization provides a clear comparison of long-term wage and employment changes for each selected occupation category

Visualization H2A



- 1. Geographic roles:** City to latitude and longitude fields, allowing the map view.
- 2. Columns/Rows:**
 - Longitude (generated) → Columns
 - Latitude (generated) → Rows
- 3. Marks card:**
 - Circle was selected as the mark type
 - Color: Wage to produce the red → blue gradient
 - Size: Number of Employed to scale bubble size by employment demand
 - Label: City names and wage values
- 4. Filtering:**
 - Year: 2025
 - Number of Employed
- 5. Tooltip customization:** Wages and employment values were formatted for readability (e.g., "\$85K").

Tableau Dashboard

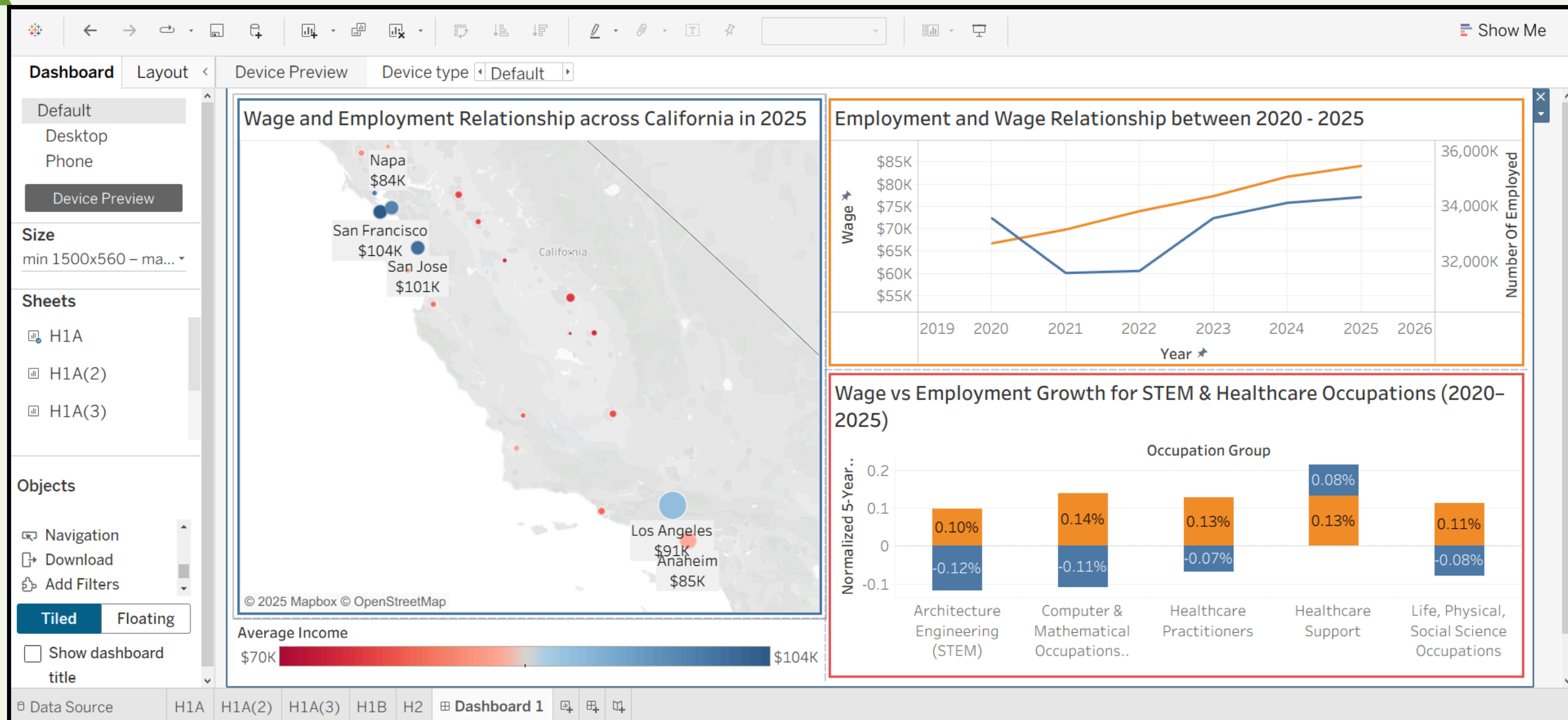




Tableau Dashboard

- The top-right visual shows that from 2020–2022, there is no clear correlation between employment and wages, largely due to COVID-19 and other external disruptions.
- From 2023–2025, a more consistent pattern appears, with employment and wages increasing in parallel, reflecting a return to typical labor-market conditions.
- The bottom-right visual highlights that only occupations with rising employment also experience wage growth. Within STEM, demand is strongest in science—especially healthcare, which drives this trend.
- Across California, regions with the highest wages (e.g., San Francisco, San Jose) do not show notable employment growth, likely due to saturation in tech sectors.
- In contrast, areas like Los Angeles show higher employment levels, but fall into a mid-to-lower wage range.
- Overall takeaway: Occupational demand significantly shapes employment distribution and wage competitiveness. Aligning wages with demand is critical for attracting and retaining long-term talent, particularly in high-growth fields such as healthcare.

